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Chapter 28 - Frobenius Series

Chapter 27 built ordinary power-series solutions around ordinary points. At a singular point, the same assumption can fail because the solution may begin with a fractional or negative power rather than a constant term.

The method of Frobenius repairs the series by allowing an unknown leading power:

$$y(x) = x^r \sum_{n=0}^{\infty} a_n x^n.$$

This chapter will:

- distinguish ordinary, regular singular, and irregular singular points
- explain why the extra exponent r is needed
- derive the indicial equation from the lowest power
- construct two solutions when the indicial roots are suitably separated
- explain repeated roots, integer differences, logarithmic terms, and complex roots
- translate a regular singular point away from the origin
- build a reliable Frobenius workflow with locally checkable recurrences

Classifying A Singular Point

The Goal Of This Block

The goal is to decide whether a singular point is mild enough for the Frobenius method.

a regular singular point permits controlled poles; an irregular singular point is more severe.

Start From Normalized Form

Consider the homogeneous second-order equation:

$$b_2(x)y'' + b_1(x)y' + b_0(x)y = 0.$$

Where $b_2(x) \neq 0$, divide by $b_2(x)$:

$$y'' + P(x)y' + Q(x)y = 0,$$

with:

$$P(x) = \frac{b_1(x)}{b_2(x)}, \quad Q(x) = \frac{b_0(x)}{b_2(x)}.$$

Point classification must use P and Q after this normalization.

Three Possible Point Types

At $x = x_0$:

1. The point is **ordinary** if P and Q are analytic at x_0 .
2. It is **regular singular** if it is not ordinary, but both $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at x_0 .
3. It is **irregular singular** if it is singular and at least one of those two corrected functions is still not analytic at x_0 .

At the origin, the regular-singular test becomes:

$$xP(x) \text{ and } x^2Q(x) \text{ must both be analytic at } 0.$$

This means P may have at most a first-order pole and Q may have at most a second-order pole.

A Regular Singular Example

Classify $x = 0$ for:

$$x^2y'' + x(1+x)y' + (2+x)y = 0.$$

For $x \neq 0$, divide by x^2 :

$$y'' + \frac{1+x}{x}y' + \frac{2+x}{x^2}y = 0.$$

Thus:

$$P(x) = \frac{1+x}{x}, \quad Q(x) = \frac{2+x}{x^2}.$$

Neither function is analytic at 0, so the point is singular. Now apply the regular-singular corrections:

$$\begin{aligned} xP(x) &= x \left(\frac{1+x}{x} \right) \\ &= 1+x, \end{aligned}$$

and:

$$\begin{aligned} x^2Q(x) &= x^2 \left(\frac{2+x}{x^2} \right) \\ &= 2+x. \end{aligned}$$

Both corrected functions are polynomials, so:

$x = 0$ is a regular singular point.

An Irregular Singular Example

Classify $x = 0$ for:

$$x^3y'' + (1+x)y' + y = 0.$$

For $x \neq 0$, divide by x^3 :

$$y'' + \frac{1+x}{x^3}y' + \frac{1}{x^3}y = 0.$$

Thus:

$$P(x) = \frac{1+x}{x^3}, \quad Q(x) = \frac{1}{x^3}.$$

For this equation, $P(x) = (1+x)/x^3$. Multiply it by one power of x :

$$xP(x) = \frac{1+x}{x^2}.$$

This still has a pole at 0, so it is not analytic there. The regular singular test fails, and:

$$x = 0 \text{ is an irregular singular point.}$$

It is unnecessary to test $x^2Q(x)$ once one required condition has failed, although here $x^2Q(x) = 1/x$ also fails.

Ordinary Does Not Mean Regular Singular

For:

$$(1+x)y'' + xy' + 3y = 0,$$

the leading coefficient is 1 at $x = 0$. After normalization:

$$P(x) = \frac{x}{1+x}, \quad Q(x) = \frac{3}{1+x}.$$

Both are analytic at 0, so the point is ordinary. We do not relabel an ordinary point as regular singular merely because xP and x^2Q are also analytic.

Block 1 Summary

Normalize first. If the point is singular, test $(x-x_0)P(x)$ and $(x-x_0)^2Q(x)$. Frobenius theory applies to the regular singular case.

The Frobenius Form And Its Derivatives

The Goal Of This Block

The goal is to see exactly how the Frobenius assumption differs from an ordinary power series and to derive its derivative formulas.

Why An Ordinary Series May Be Too Restrictive

The Euler equation:

$$x^2 y'' - \frac{1}{2} x y' - \frac{3}{2} y = 0$$

has solutions $x^{3/2}$ and x^{-1} . Neither solution is an ordinary Taylor series beginning with $x^0, x^1, x^2, \dots$ at the origin.

The missing flexibility is the leading exponent. Frobenius introduces an unknown r so the series can begin at x^r .

The Frobenius Assumption

Assume:

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r \sum_{n=0}^{\infty} a_n x^n,$$

where:

- r is an exponent to be determined
- $a_0 \neq 0$ so that x^r is genuinely the leading power
- $a_1, a_2, \dots$ are coefficients to be determined

Written term by term:

$$y = a_0x^r + a_1x^{r+1} + a_2x^{r+2} + \dots$$

If $r = 0$, this reduces to an ordinary power series.

Differentiate Term By Term

Differentiate a_nx^{n+r} :

$$\frac{d}{dx}(a_nx^{n+r}) = (n+r)a_nx^{n+r-1}.$$

Therefore:

$$y' = \sum_{n=0}^{\infty} (n+r)a_nx^{n+r-1}.$$

Differentiate one term of y' again:

$$\frac{d}{dx}[(n+r)a_nx^{n+r-1}] = (n+r)(n+r-1)a_nx^{n+r-2}.$$

Hence:

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)a_nx^{n+r-2}.$$

Check The Leading Terms

From the expanded Frobenius series:

$$y = a_0x^r + a_1x^{r+1} + a_2x^{r+2} + \dots,$$

the derivatives begin:

$$\begin{aligned} y' &= ra_0x^{r-1} + (r+1)a_1x^r + (r+2)a_2x^{r+1} + \dots, \\ y'' &= r(r-1)a_0x^{r-2} + r(r+1)a_1x^{r-1} \\ &\quad + (r+1)(r+2)a_2x^r + \dots. \end{aligned}$$

Setting $n = 0, 1, 2$ in the summation formulas reproduces these terms.

The Natural Solution Interval

If r is fractional or complex, x^r requires a branch choice. To keep the real-valued discussion unambiguous, we usually work on:

$$0 < x < R.$$

The singular centre itself need not belong to the solution interval. The series describes behaviour as x approaches the centre from within the interval.¹

Block 2 Summary

Frobenius replaces integer powers x^n by shifted powers x^{n+r} . The leading coefficient must satisfy $a_0 \neq 0$, while the equation determines r and the later coefficients.

¹ Solutions on $x < 0$ can also be studied, but fractional powers require additional sign or branch conventions. Restricting to $x > 0$ lets us use $x^r = e^{r \ln x}$ consistently.

The Indicial Equation And General Recurrence Shape

The Goal Of This Block

The goal is to identify the special lowest-power equation that determines r , and separate it from the recurrence for later coefficients.

Rewrite A Regular-Singular Equation

At a regular singular point $x = 0$, write the normalized equation as:

$$y'' + P(x)y' + Q(x)y = 0.$$

Define analytic functions:

$$p(x) = xP(x), \quad q(x) = x^2Q(x).$$

Then multiply the differential equation by x^2 :

$$\boxed{x^2y'' + xp(x)y' + q(x)y = 0}.$$

Because p and q are analytic at 0, write:

$$p(x) = \sum_{k=0}^{\infty} p_k x^k, \quad q(x) = \sum_{k=0}^{\infty} q_k x^k.$$

Find The Lowest Power

For the Frobenius series:

$$y = \sum_{n=0}^{\infty} a_n x^{n+r},$$

the three leading contributions are:

$$\begin{aligned} x^2 y'' &\supset r(r-1)a_0 x^r, \\ xp(x)y' &\supset p_0 r a_0 x^r, \\ q(x)y &\supset q_0 a_0 x^r. \end{aligned}$$

Therefore the coefficient of the lowest power x^r is:

$$[r(r-1) + p_0 r + q_0] a_0.$$

Set it equal to zero:

$$[r(r-1) + p_0 r + q_0] a_0 = 0.$$

Since $a_0 \neq 0$, divide by a_0 :

$$\boxed{r(r-1) + p_0 r + q_0 = 0}.$$

This quadratic equation is the **indicial equation**.

Why We Do Not Choose A Zero

The lowest-power equation is a product:

$$I(r)a_0 = 0,$$

where:

$$I(r) = r(r - 1) + p_0r + q_0.$$

Choosing $a_0 = 0$ would remove the assumed leading term and make a later coefficient the true starting coefficient. Frobenius instead keeps $a_0 \neq 0$ and requires:

$$I(r) = 0.$$

The roots of $I(r)$ are the indicial roots.

The General Later-Coefficient Equation

Let:

$$I(s) = s(s - 1) + p_0s + q_0.$$

After the lowest power, the coefficient of x^{n+r} for $n \geq 1$ has the form:

$$I(n+r)a_n + \sum_{k=1}^n [p_k(n-k+r) + q_k] a_{n-k} = 0.$$

The first term contains the new coefficient a_n . The finite sum contains earlier coefficients $a_0, \dots, a_{n-1}$.

Before isolating a_n , keep the complete coefficient equation in view:

$$I(n+r)a_n + \sum_{k=1}^n [p_k(n-k+r) + q_k] a_{n-k} = 0.$$

When $I(n+r) \neq 0$, subtract the finite sum and divide by $I(n+r)$:

$$a_n = -\frac{\sum_{k=1}^n [p_k(n-k+r) + q_k] a_{n-k}}{I(n+r)}.$$

This general formula is useful for structure, but in worked examples it is usually clearer to substitute the specific series and simplify directly.

The Denominator Warning

The recurrence requires division by $I(n+r)$. If:

$$I(n+r) = 0$$

for some positive integer n , the recurrence may fail to determine a_n . This is the obstruction that can occur when the two indicial roots differ by an integer.

Do not cancel or divide by a recurrence factor until checking whether it can be zero for the chosen root and allowed indices.

Block 3 Summary

The coefficient of the lowest power gives the indicial equation. Later powers give a recurrence, whose denominator must be checked for zero before it is used.

Two Distinct Roots With A Noninteger Difference

The Goal Of This Block

The goal is to construct two Frobenius solutions when both indicial roots can be used without an obstruction.

The Problem And Point Check

Solve near $x = 0$ on $x > 0$:

$$x^2 y'' + \frac{1}{2} x y' + \left(x - \frac{3}{2}\right) y = 0.$$

After division by x^2 :

$$y'' + \frac{1}{2x} y' + \left(\frac{1}{x} - \frac{3}{2x^2}\right) y = 0.$$

Thus:

$$P(x) = \frac{1}{2x}, \quad Q(x) = \frac{1}{x} - \frac{3}{2x^2}.$$

Apply the regular-singular corrections:

$$xP(x) = \frac{1}{2},$$

and:

$$x^2 Q(x) = x - \frac{3}{2}.$$

Both are analytic at 0, so the origin is regular singular.

Substitute The Frobenius Series

Use:

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^{n+r}, \\ y' &= \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}, \\ y'' &= \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}. \end{aligned}$$

Multiply the derivative series by their equation coefficients:

$$x^2 y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r},$$

$$\frac{1}{2} x y' = \sum_{n=0}^{\infty} \frac{1}{2} (n+r) a_n x^{n+r},$$

and:

$$-\frac{3}{2} y = \sum_{n=0}^{\infty} -\frac{3}{2} a_n x^{n+r}.$$

The remaining term is:

$$x y = \sum_{n=0}^{\infty} a_n x^{n+r+1}.$$

Reindex it by replacing the old index with $n-1$:

$$x y = \sum_{n=1}^{\infty} a_{n-1} x^{n+r}.$$

Derive The Indicial Equation

The term xy has no contribution to the lowest power x^r . The coefficient of x^r from the other three terms is:

$$\left[r(r-1) + \frac{1}{2}r - \frac{3}{2} \right] a_0.$$

Keep $a_0 \neq 0$ and set the bracket equal to zero:

$$r(r-1) + \frac{1}{2}r - \frac{3}{2} = 0.$$

Expand:

$$r^2 - \frac{1}{2}r - \frac{3}{2} = 0.$$

Factor:

$$\left(r - \frac{3}{2}\right)(r+1) = 0.$$

Therefore the indicial roots are:

$$\boxed{r_1 = \frac{3}{2}, \quad r_2 = -1}.$$

Their difference is:

$$r_1 - r_2 = \frac{3}{2} - (-1) = \frac{5}{2},$$

which is not an integer. We expect two independent Frobenius series.

Derive The Recurrence

For $n \geq 1$, the coefficient of x^{n+r} is:

$$\left[(n+r)(n+r-1) + \frac{1}{2}(n+r) - \frac{3}{2} \right] a_n + a_{n-1} = 0.$$

Define:

$$I(s) = s^2 - \frac{1}{2}s - \frac{3}{2} = \left(s - \frac{3}{2}\right)(s + 1).$$

Then the recurrence is:

$$I(n + r)a_n + a_{n-1} = 0.$$

Solve for a_n :

$$a_n = -\frac{a_{n-1}}{I(n + r)}, \quad n \geq 1.$$

Use The Larger Root

Set $r = 3/2$. Then:

$$\begin{aligned} I\left(n + \frac{3}{2}\right) &= \left(n + \frac{3}{2} - \frac{3}{2}\right)\left(n + \frac{3}{2} + 1\right) \\ &= n\left(n + \frac{5}{2}\right). \end{aligned}$$

Therefore:

$$a_n = -\frac{a_{n-1}}{n(n + 5/2)}.$$

Set $n = 1$:

$$\begin{aligned} a_1 &= -\frac{a_0}{1(7/2)} \\ &= -\frac{2}{7}a_0. \end{aligned}$$

For the next coefficient, return to $a_n = -a_{n-1}/[n(n + 5/2)]$ and set $n = 2$:

$$\begin{aligned} a_2 &= -\frac{a_1}{2(9/2)} \\ &= -\frac{-2a_0/7}{9} \\ &= \frac{2}{63}a_0. \end{aligned}$$

The first solution is:

$$y_1(x) = x^{3/2} \left(1 - \frac{2}{7}x + \frac{2}{63}x^2 + \cdots \right).$$

The scale a_0 has been absorbed into the arbitrary constant that will multiply y_1 .

Use The Smaller Root

Set $r = -1$. Then:

$$\begin{aligned} I(n-1) &= \left(n-1-\frac{3}{2}\right)(n-1+1) \\ &= n\left(n-\frac{5}{2}\right). \end{aligned}$$

Therefore:

$$a_n = -\frac{a_{n-1}}{n(n-5/2)}.$$

Set $n = 1$:

$$\begin{aligned} a_1 &= -\frac{a_0}{1(-3/2)} \\ &= \frac{2}{3}a_0. \end{aligned}$$

Set $n = 2$:

$$\begin{aligned} a_2 &= -\frac{a_1}{2(-1/2)} \\ &= a_1 \\ &= \frac{2}{3}a_0. \end{aligned}$$

Set $n = 3$:

$$\begin{aligned} a_3 &= -\frac{a_2}{3(1/2)} \\ &= -\frac{2}{3}a_2 \\ &= -\frac{4}{9}a_0. \end{aligned}$$

The second solution is:

$$y_2(x) = x^{-1} \left(1 + \frac{2}{3}x + \frac{2}{3}x^2 - \frac{4}{9}x^3 + \dots \right).$$

Assemble The General Solution

The two indicial roots differ by a noninteger, so both Frobenius solutions are available and linearly independent. Therefore:

$$y(x) = c_1 x^{3/2} \left(1 - \frac{2}{7}x + \frac{2}{63}x^2 + \dots \right) + c_2 x^{-1} \left(1 + \frac{2}{3}x + \frac{2}{3}x^2 - \frac{4}{9}x^3 + \dots \right), \quad x > 0.$$

The two leading behaviours, $x^{3/2}$ and x^{-1} , are already different enough to show that neither solution is a constant multiple of the other.

Block 4 Summary

When distinct real indicial roots differ by a noninteger, substitute each root into the recurrence. Each root produces one Frobenius series, and their linear combination gives the general solution.

Repeated Roots And Logarithmic Partners

The Goal Of This Block

The goal is to understand why one repeated indicial root may produce only one ordinary Frobenius series and why a logarithm supplies the missing independent solution.

A Repeated-Root Equation

Solve on $x > 0$:

$$x^2 y'' - 3xy' + 4y = 0.$$

After normalization:

$$y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0.$$

Thus:

$$xP(x) = -3, \quad x^2Q(x) = 4.$$

Both corrected functions are analytic, so $x = 0$ is regular singular.

Find The Indicial Equation

Substitute:

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}.$$

The three equation terms become:

$$\begin{aligned}x^2 y'' &= \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r}, \\ -3xy' &= \sum_{n=0}^{\infty} -3(n+r) a_n x^{n+r}, \\ 4y &= \sum_{n=0}^{\infty} 4a_n x^{n+r}.\end{aligned}$$

Combine the coefficients:

$$\sum_{n=0}^{\infty} [(n+r)(n+r-1) - 3(n+r) + 4] a_n x^{n+r} = 0.$$

For the lowest power, set $n = 0$:

$$[r(r-1) - 3r + 4] a_0 = 0.$$

Since $a_0 \neq 0$:

$$r(r-1) - 3r + 4 = 0.$$

Simplify and factor:

$$\begin{aligned}r^2 - 4r + 4 &= 0, \\ (r-2)^2 &= 0.\end{aligned}$$

The indicial root is repeated:

$$\boxed{r_1 = r_2 = 2}.$$

Find The First Solution

The coefficient bracket can be rewritten as:

$$(n + r - 2)^2.$$

Set $r = 2$. For every $n \geq 1$:

$$n^2 a_n = 0.$$

Since $n^2 \neq 0$ for $n \geq 1$:

$$a_n = 0, \quad n \geq 1.$$

Only the leading coefficient remains:

$$y_1(x) = a_0 x^2.$$

Absorb a_0 into the eventual arbitrary constant and choose:

$$\boxed{y_1(x) = x^2}.$$

Where The Logarithm Comes From

For $x > 0$:

$$x^r = e^{r \ln x}.$$

Differentiate with respect to the exponent r while treating x as fixed:

$$\frac{\partial}{\partial r} x^r = x^r \ln x.$$

When two indicial roots merge, the second power x^{r_2} no longer gives a new direction. Differentiation with respect to the root parameter produces a logarithmic partner. In this example:

$$y_2(x) = x^2 \ln x.$$

This parameter explanation is intuition for the logarithm; the next section verifies the proposed function directly.

Verify The Logarithmic Solution

Let:

$$y_2 = x^2 \ln x.$$

Use the product rule:

$$\begin{aligned} y_2' &= 2x \ln x + x^2 \left(\frac{1}{x} \right) \\ &= 2x \ln x + x. \end{aligned}$$

Differentiate again:

$$\begin{aligned} y_2'' &= 2 \ln x + 2x \left(\frac{1}{x} \right) + 1 \\ &= 2 \ln x + 3. \end{aligned}$$

Using $y_2' = 2x \ln x + x$ and $y_2'' = 2 \ln x + 3$, substitute into the differential equation:

$$\begin{aligned} x^2 y_2'' - 3x y_2' + 4y_2 &= x^2(2 \ln x + 3) - 3x(2x \ln x + x) + 4x^2 \ln x \\ &= (2 - 6 + 4)x^2 \ln x + (3 - 3)x^2 \\ &= 0. \end{aligned}$$

Thus $x^2 \ln x$ is a solution.

Check Linear Independence

Use:

$$y_1 = x^2, \quad y_2 = x^2 \ln x.$$

Their derivatives are:

$$y_1' = 2x, \quad y_2' = 2x \ln x + x.$$

Calculate the Wronskian:

$$\begin{aligned} W[y_1, y_2] &= y_1 y_2' - y_2 y_1' \\ &= x^2(2x \ln x + x) - x^2 \ln x(2x) \\ &= 2x^3 \ln x + x^3 - 2x^3 \ln x \\ &= x^3. \end{aligned}$$

For $x > 0$, $x^3 \neq 0$, so the solutions are linearly independent.

The General Solution

The two independent solutions are x^2 and $x^2 \ln x$. Therefore:

$$y(x) = c_1 x^2 + c_2 x^2 \ln x, \quad x > 0.$$

The logarithm is not an arbitrary decoration. It replaces the second power solution lost when the indicial roots coincide.

Block 5 Summary

A repeated indicial root usually gives one direct Frobenius series. A second solution has the general shape:

$$y_2 = y_1 \ln x + x^r \sum_{n=0}^{\infty} b_n x^n,$$

although the additional series may vanish in a simple Euler equation.

Integer Differences And Recurrence Obstructions

The Goal Of This Block

The goal is to distinguish two facts that are often incorrectly merged:

- an integer root difference creates the possibility of a logarithm
- it does not force a logarithm in every equation

An Integer Difference With No Obstruction

Consider:

$$x^2 y'' - x y' = 0.$$

Substituting a leading power $y = x^r$ gives:

$$\begin{aligned}x^2[r(r-1)x^{r-2}] - x[rx^{r-1}] &= 0, \\[r(r-1) - r]x^r &= 0.\end{aligned}$$

The indicial equation is:

$$r(r-1) - r = 0.$$

Factor:

$$r(r-2) = 0.$$

Thus:

$$r_1 = 2, \quad r_2 = 0.$$

The roots differ by the integer 2, yet both power solutions work:

$$y_1 = x^2, \quad y_2 = 1.$$

Therefore:

$$\boxed{y = c_1 x^2 + c_2}.$$

No logarithm is needed because the lower-root construction does not fail.

An Equation With An Obstructed Lower Root

Consider:

$$x^2 y'' + xy' + (9x^2 - 1)y = 0.$$

After normalization:

$$P(x) = \frac{1}{x}, \quad Q(x) = 9 - \frac{1}{x^2}.$$

Since:

$$xP(x) = 1, \quad x^2Q(x) = 9x^2 - 1,$$

the origin is regular singular.

Derive The Indicial Equation And Recurrence

For the Frobenius series, the terms $x^2 y'' + xy' - y$ combine as:

$$\sum_{n=0}^{\infty} [(n+r)^2 - 1] a_n x^{n+r}.$$

The remaining term is:

$$9x^2 y = 9 \sum_{n=0}^{\infty} a_n x^{n+r+2} = 9 \sum_{n=2}^{\infty} a_{n-2} x^{n+r}.$$

The lowest-power equation is:

$$(r^2 - 1)a_0 = 0.$$

Since $a_0 \neq 0$, the indicial equation is:

$$r^2 - 1 = 0.$$

Thus:

$$\boxed{r_1 = 1, \quad r_2 = -1}.$$

The coefficient of x^{r+1} gives:

$$[(r+1)^2 - 1]a_1 = 0.$$

For $n \geq 2$:

$$\boxed{[(n+r)^2 - 1]a_n + 9a_{n-2} = 0}.$$

Construct The Larger-Root Solution

Set $r = 1$. The $n = 1$ equation becomes:

$$[(2)^2 - 1]a_1 = 0,$$

so:

$$3a_1 = 0, \quad a_1 = 0.$$

For $n \geq 2$, substitute $r = 1$:

$$[(n+1)^2 - 1]a_n + 9a_{n-2} = 0.$$

Factor the bracket:

$$(n+1)^2 - 1 = n(n+2).$$

Therefore:

$$a_n = -\frac{9}{n(n+2)}a_{n-2}.$$

Set $n = 2$:

$$\begin{aligned}a_2 &= -\frac{9}{(2)(4)}a_0 \\ &= -\frac{9}{8}a_0.\end{aligned}$$

Set $n = 4$:

$$\begin{aligned}a_4 &= -\frac{9}{(4)(6)}a_2 \\ &= -\frac{3}{8}\left(-\frac{9}{8}a_0\right) \\ &= \frac{27}{64}a_0.\end{aligned}$$

Thus one solution is:

$$y_1(x) = x \left(1 - \frac{9}{8}x^2 + \frac{27}{64}x^4 + \dots \right).$$

Watch The Smaller Root Fail Locally

Now set $r = -1$. The $n = 1$ equation is:

$$[(0)^2 - 1]a_1 = 0,$$

so:

$$-a_1 = 0, \quad a_1 = 0.$$

Next use the $n = 2$ equation:

$$[(2 - 1)^2 - 1]a_2 + 9a_0 = 0.$$

Simplify its coefficient of a_2 :

$$[(1)^2 - 1]a_2 + 9a_0 = 0,$$

so:

$$0 \cdot a_2 + 9a_0 = 0.$$

This would require $a_0 = 0$, contradicting the Frobenius assumption $a_0 \neq 0$. The lower root therefore does not produce a second pure Frobenius series.

The second solution instead has the structure:

$$y_2(x) = Cy_1(x) \ln x + x^{-1} \sum_{n=0}^{\infty} b_n x^n.$$

The constant C and coefficients b_n require an extended calculation. The important diagnosis here is the exact line at which the pure recurrence becomes inconsistent.

The Integer-Difference Decision

If $r_1 - r_2 = N$ is a positive integer:

1. Use the larger root r_1 first; it produces a Frobenius solution.
2. Try the smaller root r_2 in the recurrence.
3. If every required division remains valid, the smaller root gives a second pure Frobenius solution.
4. If a denominator vanishes and leaves an inconsistent equation, the second solution contains a logarithmic term.

Do not decide whether a logarithm appears from the root difference alone.

Block 6 Summary

An integer difference is a warning to inspect the smaller-root recurrence. A zero denominator may be harmless, may leave a free coefficient, or may create an obstruction. The actual coefficient equation determines the outcome.

Complex Roots And Shifted Singular Centres

The Goal Of This Block

The goal is to interpret complex indicial roots as real oscillations in $\ln x$ and translate a regular singular point that is not at the origin.

A Complex-Root Euler Equation

Consider:

$$x^2 y'' + xy' + 4y = 0, \quad x > 0.$$

Try $y = x^r$. Then:

$$\begin{aligned}x^2 y'' + x y' + 4y &= [r(r-1) + r + 4]x^r \\ &= (r^2 + 4)x^r.\end{aligned}$$

The indicial equation is:

$$r^2 + 4 = 0,$$

with roots:

$$r = \pm 2i.$$

Convert Complex Powers To Real Functions

For $x > 0$:

$$x^{2i} = e^{2i \ln x}.$$

Euler's formula gives:

$$e^{2i \ln x} = \cos(2 \ln x) + i \sin(2 \ln x).$$

The conjugate power x^{-2i} supplies the same real and imaginary parts. Therefore two real solutions are:

$$\cos(2 \ln x) \quad \text{and} \quad \sin(2 \ln x).$$

The general real solution is:

$$y(x) = c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x), \quad x > 0.$$

The oscillation occurs in logarithmic position rather than in x itself.

Verify One Logarithmic Oscillation

Let:

$$y = \cos(2 \ln x).$$

Differentiate using the chain rule:

$$y' = -\frac{2}{x} \sin(2 \ln x).$$

Differentiate again using the product rule:

$$y'' = \frac{2}{x^2} \sin(2 \ln x) - \frac{4}{x^2} \cos(2 \ln x).$$

Substitute:

$$\begin{aligned} x^2 y'' + x y' + 4y &= [2 \sin(2 \ln x) - 4 \cos(2 \ln x)] \\ &\quad - 2 \sin(2 \ln x) + 4 \cos(2 \ln x) \\ &= 0. \end{aligned}$$

Translate A Singular Centre

Solve near $x = 2$:

$$(x - 2)^2 y'' + 2(x - 2)y' - 2y = 0.$$

Set:

$$t = x - 2.$$

Because this is a translation, $dy/dx = dy/dt$ and $d^2y/dx^2 = d^2y/dt^2$. The equation becomes:

$$t^2 y'' + 2ty' - 2y = 0$$

near $t = 0$.

Solve And Return To X

Try $y = t^r$. Then:

$$t^2 y'' + 2ty' - 2y = [r(r - 1) + 2r - 2]t^r.$$

The indicial equation is:

$$r(r - 1) + 2r - 2 = 0.$$

Simplify and factor:

$$\begin{aligned} r^2 + r - 2 &= 0, \\ (r - 1)(r + 2) &= 0. \end{aligned}$$

Solving $(r - 1)(r + 2) = 0$ gives:

$$r_1 = 1, \quad r_2 = -2.$$

Both Euler powers work, so in the translated variable:

$$y(t) = c_1 t + c_2 t^{-2}.$$

Return to $t = x - 2$:

$$\boxed{y(x) = c_1(x - 2) + c_2(x - 2)^{-2}}.$$

For a real solution using these powers without crossing the singular point, choose an interval such as $x > 2$ or $x < 2$.

Block 7 Summary

Complex roots $\alpha \pm i\beta$ lead to real factors $x^\alpha \cos(\beta \ln x)$ and $x^\alpha \sin(\beta \ln x)$. A nonzero singular centre is translated to the origin with $t = x - x_0$ before applying the same analysis.

A Reliable Workflow And Common Mistakes

The Goal Of This Block

The goal is to keep point classification, lowest-power matching, recurrence construction, and root-case decisions in a dependable order.

The Frobenius Workflow

Step 1: Normalize and classify the centre. Calculate P and Q . If the point is singular, test $(x - x_0)P$ and $(x - x_0)^2Q$.

Step 2: Translate a nonzero centre. Use $t = x - x_0$ and rewrite every coefficient in terms of t .

Step 3: Simplify the equation before substituting. Clearing rational denominators often reduces the number of series products.

Step 4: Write all three Frobenius series. Use:

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^{n+r}, \\ y' &= \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1}, \\ y'' &= \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}. \end{aligned}$$

Step 5: Align every term to powers of x^{n+r} . Show each index shift and preserve its lower limit.

Step 6: Isolate the lowest power. Its coefficient times a_0 gives the indicial equation because $a_0 \neq 0$.

Step 7: Derive the later recurrence. State its valid range and inspect every denominator before dividing.

Step 8: Order and compare the roots. Identify whether they are distinct with noninteger difference, repeated, integer-separated, or complex conjugates.

Step 9: Generate the available solution branches. For an integer difference, use the larger root first and test the smaller root rather than assuming its outcome.

Step 10: Reconstruct, state the domain, and verify. Restore the factor x^r , return from any translated variable, and keep the solution interval away from the singular point.

Common Mistake: Setting R To Zero Automatically

Using:

$$y = \sum_{n=0}^{\infty} a_n x^n$$

at a regular singular point assumes $r = 0$ before the indicial equation has been solved. This may discard fractional, negative, or complex leading powers.

Common Mistake: Letting A Zero

The Frobenius assumption requires $a_0 \neq 0$. If the lowest-power equation is:

$$I(r)a_0 = 0,$$

choose $I(r) = 0$, not $a_0 = 0$. Otherwise the displayed leading term was not actually the leading term.

Common Mistake: Mixing The Indicial Equation With The Recurrence

The indicial equation comes from the single lowest power. The recurrence comes from later powers. Combining them too early can hide exceptional equations such as the separate $n = 1$ condition in a two-step recurrence.

Common Mistake: Dividing Before Checking The Denominator

Before using:

$$a_n = -\frac{\text{earlier coefficients}}{I(n+r)},$$

substitute the selected root and ask whether $I(n+r) = 0$ for an allowed positive integer n . A zero denominator is mathematical information, not an algebraic inconvenience to cancel away.

Common Mistake: Assuming Every Integer Difference Creates A Log

Integer-separated roots may produce two pure power or Frobenius solutions, as in $x^2y'' - xy' = 0$. A logarithm is needed when the smaller-root construction is obstructed, not merely because the difference is an integer.

Common Mistake: Dropping The Leading Power

If the coefficient series is:

$$a_0 + a_1x + a_2x^2 + \cdots,$$

the Frobenius solution is:

$$x^r(a_0 + a_1x + a_2x^2 + \cdots).$$

The factor x^r is part of every term and must remain in the final answer.

Common Mistake: Including The Singular Centre In The Interval

A formula containing x^{-1} or $\ln x$ is not defined at $x = 0$. Write a domain such as $0 < x < R$ rather than claiming the solution is valid on an interval containing the singular point.

Block 8 Summary

Classify before expanding, derive the indicial equation before the recurrence, inspect recurrence denominators before dividing, and let the relationship between the roots determine how the second solution must be sought.

Original Practice Set

Problems 1 To 4: Classification And Indicial Equations

Problem 1: Classify Three Origins. Classify $x = 0$ as ordinary, regular singular, or irregular singular for each equation.

$$(a) \quad x^2 y'' + x(2 + x)y' + (1 - x)y = 0,$$

$$(b) \quad x^4 y'' + xy' + y = 0,$$

$$(c) \quad (1 + x^2)y'' + xy' + e^x y = 0.$$

Problem 2: Classify A Shifted Point. Determine whether $x = -1$ is a regular singular point of:

$$(x + 1)^2 y'' + (x + 1)(2 - x)y' + (3 + x)y = 0.$$

Problem 3: Differentiate A Frobenius Series. Starting from:

$$y = \sum_{n=0}^{\infty} a_n x^{n+r},$$

derive y' and y'' , then write the $n = 0$ and $n = 1$ terms of each derivative.

Problem 4: Use The General Indicial Formula. For a regular-singular equation with:

$$p_0 = 3, \quad q_0 = -4,$$

find the indicial equation and its roots.

Problems 5 To 8: Real Root Cases

Problem 5: Solve A Distinct-Root Euler Equation. Solve on $x > 0$:

$$x^2 y'' + 2xy' - 6y = 0.$$

Problem 6: Generate A Larger-Root Series. For:

$$x^2 y'' + 2xy' + (x - 2)y = 0,$$

find the indicial roots and use the larger root to calculate a Frobenius solution through the term containing x^4 .

Problem 7: Solve A Repeated-Root Euler Equation. Solve on $x > 0$:

$$x^2 y'' + 5xy' + 4y = 0.$$

Problem 8: Check An Integer Difference Without A Logarithm. Find the general solution of:

$$x^2 y'' - 3xy' + 3y = 0$$

and explain why an integer difference between the roots causes no obstruction here.

Problems 9 To 12: Obstructions, Complex Roots, And Shifts

Problem 9: Locate A Lower-Root Obstruction. For:

$$x^2y'' + xy' + (4x^2 - 4)y = 0,$$

derive the indicial roots and show exactly where the recurrence for the smaller root becomes inconsistent.

Problem 10: Convert Complex Roots To Real Solutions. Solve on $x > 0$:

$$x^2y'' - xy' + 5y = 0.$$

Problem 11: Translate A Singular Centre. Solve near $x = -3$:

$$(x + 3)^2y'' - 2y = 0.$$

Problem 12: Verify A Logarithmic Pair. For:

$$x^2y'' + 3xy' + y = 0,$$

verify that $y_1 = x^{-1}$ and $y_2 = x^{-1} \ln x$ are solutions on $x > 0$, and calculate their Wronskian.

Worked Answers: Problems 1 To 4**Answer 1.**

Equation (a). For $x \neq 0$, divide by x^2 :

$$y'' + \frac{2+x}{x}y' + \frac{1-x}{x^2}y = 0.$$

Thus:

$$P(x) = \frac{2+x}{x}, \quad Q(x) = \frac{1-x}{x^2}.$$

The point is singular. Apply the corrected tests:

$$xP(x) = 2 + x, \quad x^2Q(x) = 1 - x.$$

Both are analytic at 0, so:

$x = 0$ is regular singular.

Equation (b). Divide by x^4 :

$$y'' + \frac{1}{x^3}y' + \frac{1}{x^4}y = 0.$$

Thus:

$$xP(x) = \frac{1}{x^2}, \quad x^2Q(x) = \frac{1}{x^2}.$$

Neither corrected coefficient is analytic at 0, so:

$x = 0 \text{ is irregular singular.}$

Equation (c). The leading coefficient at 0 is:

$$1 + 0^2 = 1 \neq 0.$$

After normalization:

$$P(x) = \frac{x}{1+x^2}, \quad Q(x) = \frac{e^x}{1+x^2}.$$

Both are analytic at 0, so:

$x = 0 \text{ is ordinary.}$

Answer 2. Set:

$$t = x + 1.$$

Then $x = t - 1$ and $2 - x = 3 - t$. The equation becomes:

$$t^2 y'' + t(3 - t)y' + (t + 2)y = 0.$$

For $t \neq 0$, normalize:

$$y'' + \frac{3 - t}{t}y' + \frac{t + 2}{t^2}y = 0.$$

The corrected coefficients are:

$$tP(t) = 3 - t, \quad t^2Q(t) = t + 2.$$

Both are analytic at $t = 0$. Therefore:

$x = -1$ is a regular singular point.

Answer 3. Differentiate one term:

$$\frac{d}{dx}(a_n x^{n+r}) = (n+r)a_n x^{n+r-1}.$$

Therefore:

$$y' = \sum_{n=0}^{\infty} (n+r)a_n x^{n+r-1}.$$

Its first two terms are:

$$ra_0 x^{r-1} + (r+1)a_1 x^r.$$

Differentiate again:

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)a_n x^{n+r-2}.$$

Its first two terms are:

$$r(r-1)a_0 x^{r-2} + r(r+1)a_1 x^{r-1}.$$

Answer 4. The general indicial equation is:

$$r(r-1) + p_0r + q_0 = 0.$$

Substitute $p_0 = 3$ and $q_0 = -4$:

$$r(r-1) + 3r - 4 = 0.$$

Expand:

$$r^2 + 2r - 4 = 0.$$

Use the quadratic formula:

$$\begin{aligned} r &= \frac{-2 \pm \sqrt{2^2 - 4(1)(-4)}}{2} \\ &= \frac{-2 \pm \sqrt{20}}{2} \\ &= -1 \pm \sqrt{5}. \end{aligned}$$

Thus:

$$\boxed{r_1 = -1 + \sqrt{5}, \quad r_2 = -1 - \sqrt{5}}.$$

Worked Answers: Problems 5 To 8

Answer 5. Try $y = x^r$. Then:

$$y' = rx^{r-1}, \quad y'' = r(r-1)x^{r-2}.$$

Substitute:

$$x^2y'' + 2xy' - 6y = [r(r-1) + 2r - 6]x^r.$$

The indicial equation is:

$$r(r-1) + 2r - 6 = 0.$$

Simplify and factor:

$$\begin{aligned} r^2 + r - 6 &= 0, \\ (r-2)(r+3) &= 0. \end{aligned}$$

Thus:

$$r_1 = 2, \quad r_2 = -3.$$

Both Euler powers solve the equation, so:

$$\boxed{y(x) = c_1x^2 + c_2x^{-3}, \quad x > 0}.$$

Answer 6. Use:

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}.$$

The terms $x^2 y'' + 2xy' - 2y$ contribute:

$$\sum_{n=0}^{\infty} [(n+r)(n+r-1) + 2(n+r) - 2] a_n x^{n+r}.$$

The remaining term is:

$$xy = \sum_{n=1}^{\infty} a_{n-1} x^{n+r}.$$

The lowest-power equation gives:

$$[r(r-1) + 2r - 2]a_0 = 0.$$

Since $a_0 \neq 0$:

$$r^2 + r - 2 = 0.$$

Factor:

$$(r-1)(r+2) = 0.$$

Thus:

$$r_1 = 1, \quad r_2 = -2.$$

Matching the coefficient of x^{n+r} gives, for $n \geq 1$:

$$[(n+r-1)(n+r+2)]a_n + a_{n-1} = 0.$$

Use the larger root $r = 1$:

$$n(n+3)a_n + a_{n-1} = 0.$$

Therefore:

$$a_n = -\frac{a_{n-1}}{n(n+3)}.$$

Choose $a_0 = 1$. For $n = 1$:

$$a_1 = -\frac{1}{(1)(4)} = -\frac{1}{4}.$$

For $n = 2$:

$$\begin{aligned} a_2 &= -\frac{a_1}{(2)(5)} \\ &= \frac{1}{40}. \end{aligned}$$

For $n = 3$:

$$\begin{aligned} a_3 &= -\frac{a_2}{(3)(6)} \\ &= -\frac{1}{720}. \end{aligned}$$

Since the leading factor is $x^r = x$, the requested solution through x^4 is:

$$y_1(x) = x - \frac{1}{4}x^2 + \frac{1}{40}x^3 - \frac{1}{720}x^4 + \dots$$

Answer 7. Try $y = x^r$. The indicial equation is:

$$r(r-1) + 5r + 4 = 0.$$

Simplify:

$$r^2 + 4r + 4 = 0.$$

Factor:

$$(r+2)^2 = 0.$$

The repeated root is $r = -2$. One solution is:

$$y_1 = x^{-2}.$$

For a repeated Euler root, a logarithmic partner is:

$$y_2 = x^{-2} \ln x.$$

Therefore:

$$y(x) = c_1 x^{-2} + c_2 x^{-2} \ln x, \quad x > 0.$$

Answer 8. Try $y = x^r$. The indicial equation is:

$$r(r - 1) - 3r + 3 = 0.$$

Simplify and factor:

$$\begin{aligned} r^2 - 4r + 3 &= 0, \\ (r - 3)(r - 1) &= 0. \end{aligned}$$

Thus:

$$r_1 = 3, \quad r_2 = 1.$$

Both power functions can be checked directly:

$$y_1 = x^3, \quad y_2 = x.$$

They are not constant multiples, so:

$$\boxed{y = c_1 x^3 + c_2 x}.$$

The roots differ by 2, but no recurrence obstruction occurs because this Euler equation accepts both pure powers directly.

Worked Answers: Problems 9 To 12

Answer 9. For:

$$x^2y'' + xy' - 4y + 4x^2y = 0,$$

the Frobenius substitution gives the lowest-power equation:

$$(r^2 - 4)a_0 = 0.$$

Since $a_0 \neq 0$:

$$r^2 - 4 = 0.$$

Thus:

$$r_1 = 2, \quad r_2 = -2.$$

For $n = 1$:

$$[(r + 1)^2 - 4]a_1 = 0.$$

For $n \geq 2$:

$$\boxed{[(n + r)^2 - 4]a_n + 4a_{n-2} = 0}.$$

Now choose the smaller root $r = -2$. At $n = 1$:

$$[(-1)^2 - 4]a_1 = 0,$$

so:

$$-3a_1 = 0, \quad a_1 = 0.$$

The first step gave $a_1 = 0$. Continue with the recurrence at $n = 2$:

$$[(0)^2 - 4]a_2 + 4a_0 = 0.$$

Therefore:

$$-4a_2 + 4a_0 = 0, \quad a_2 = a_0.$$

At $n = 3$:

$$[(1)^2 - 4]a_3 + 4a_1 = 0.$$

Since $a_1 = 0$:

$$-3a_3 = 0, \quad a_3 = 0.$$

At $n = 4$:

$$[(2)^2 - 4]a_4 + 4a_2 = 0.$$

The coefficient of a_4 vanishes:

$$0 \cdot a_4 + 4a_2 = 0.$$

But $a_2 = a_0 \neq 0$, so this equation is impossible. The lower-root pure Frobenius series is obstructed at:

$$\boxed{n = 4}.$$

Answer 10. Try $y = x^r$. Substitution gives:

$$[r(r-1) - r + 5]x^r = 0.$$

The indicial equation is:

$$r^2 - 2r + 5 = 0.$$

Use the quadratic formula:

$$\begin{aligned} r &= \frac{2 \pm \sqrt{(-2)^2 - 4(1)(5)}}{2} \\ &= \frac{2 \pm \sqrt{-16}}{2} \\ &= 1 \pm 2i. \end{aligned}$$

For $r = 1 \pm 2i$:

$$x^r = x^{1 \pm 2i} = xe^{\pm 2i \ln x}.$$

Taking real and imaginary parts gives:

$$x \cos(2 \ln x) \quad \text{and} \quad x \sin(2 \ln x).$$

Therefore:

$$\boxed{y(x) = c_1 x \cos(2 \ln x) + c_2 x \sin(2 \ln x), \quad x > 0}.$$

Answer 11. Set:

$$t = x + 3.$$

The equation becomes:

$$t^2 y'' - 2y = 0.$$

Try $y = t^r$. The indicial equation is:

$$r(r - 1) - 2 = 0.$$

Factor:

$$(r - 2)(r + 1) = 0.$$

Thus:

$$r_1 = 2, \quad r_2 = -1.$$

In the translated variable:

$$y(t) = c_1 t^2 + c_2 t^{-1}.$$

Return to $t = x + 3$:

$$\boxed{y(x) = c_1 (x + 3)^2 + c_2 (x + 3)^{-1}}.$$

The singular point $x = -3$ must be excluded from the solution interval.

Answer 12. For $y_1 = x^{-1}$:

$$y_1' = -x^{-2}, \quad y_1'' = 2x^{-3}.$$

Substitute:

$$\begin{aligned} x^2 y_1'' + 3x y_1' + y_1 &= x^2(2x^{-3}) + 3x(-x^{-2}) + x^{-1} \\ &= (2 - 3 + 1)x^{-1} \\ &= 0. \end{aligned}$$

For $y_2 = x^{-1} \ln x$, use the product rule:

$$\begin{aligned} y_2' &= -x^{-2} \ln x + x^{-1} \left(\frac{1}{x} \right) \\ &= x^{-2}(1 - \ln x). \end{aligned}$$

Differentiate again:

$$\begin{aligned} y_2'' &= -2x^{-3}(1 - \ln x) - x^{-3} \\ &= x^{-3}(2 \ln x - 3). \end{aligned}$$

Substitute:

$$\begin{aligned} x^2 y_2'' + 3x y_2' + y_2 &= x^{-1}(2 \ln x - 3) + 3x^{-1}(1 - \ln x) + x^{-1} \ln x \\ &= x^{-1}[(2 - 3 + 1) \ln x + (-3 + 3)] \\ &= 0. \end{aligned}$$

Now calculate the Wronskian:

$$\begin{aligned} W[y_1, y_2] &= y_1 y_2' - y_2 y_1' \\ &= x^{-1} [x^{-2}(1 - \ln x)] - (x^{-1} \ln x)(-x^{-2}) \\ &= x^{-3}(1 - \ln x) + x^{-3} \ln x \\ &= x^{-3}. \end{aligned}$$

Since $x^{-3} \neq 0$ for $x > 0$, the two solutions are linearly independent.

Chapter 28 Final Summary

At a regular singular point $x = 0$, seek:

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}, \quad a_0 \neq 0.$$

The lowest power gives the indicial equation. Later powers give the coefficient recurrence. The relationship between the indicial roots then controls the second solution:

- a noninteger difference gives two Frobenius series
- a repeated root introduces a logarithmic partner
- an integer difference requires testing the smaller-root recurrence
- complex roots produce oscillations in $\ln x$

The dependable chain is:

classify the point $\rightarrow$ write the Frobenius series
 $\rightarrow$ isolate the lowest power
 $\rightarrow$ solve the indicial equation
 $\rightarrow$ derive and test the recurrence
 $\rightarrow$ choose the correct root case.

the indicial roots suggest the possible solution shapes, but the recurrence reveals whether those shapes can actually be completed.